

Geometry and Volume of Stepping-Stone Diversion Neighborhoods in Euclidean Space

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ABSTRACT

We develop a dimension-independent geometric, graph-theoretic, and probabilistic analysis of the stepping-stone diversion neighborhood in $\mathbb{R}^d$, for every $d \geq 2$. Its volume factorizes as the d th power of the distance between the defining sites times a normalized constant $a_{d,SS}(\alpha)$. An explicit boundary parametrization reduces the original d -dimensional volume problem to a one-dimensional integral. We recover the degenerate and Gabriel region cases and prove, in every dimension, that, after removing the two defining sites, the finite-parameter neighborhoods converge from within to the open relative-neighborhood lune. Consequently, their volumes converge to the classical unit-lune volume of the relative-neighborhood graph, for which we record an equivalent regularized incomplete-beta representation. For arbitrary ambient dimension, we also prove strict region and volume monotonicity, derive the corresponding inclusions among classical proximity graphs, establish connectivity, and show that, for $\alpha \geq 2$, the defining empty-region condition need only be evaluated for Gabriel edges that are not open-lune relative-neighborhood edges. Devroye's expected-size framework then gives the asymptotic expected degree in terms of $a_{d,SS}(\alpha)$, again for every $d \geq 2$. As an explicitly planar specialization, Watanabe's restricted-search approximation gives the corresponding graph-edge-length density. Finally, we derive a numerically stable quadrature formula and verify it independently in 18 rejection-sampling experiments, each with 10^8 samples and with every discrepancy below two Monte Carlo standard errors.

1. Introduction

An empty-region proximity graph joins two sites whenever a region determined by them contains no other site. Classical examples include the Gabriel graph, the relative-neighborhood graph, and the β -skeletons (Gabriel and Sokal, 1969; Toussaint, 1980; Jaromczyk and Toussaint, 1992; Kirkpatrick and Radke, 1985). The theoretical, probabilistic, and algorithmic properties of proximity graphs have been extensively studied (Bose and Smid, 2013; Devroye, 1988; Cardinal, Collette and Langerman, 2009). Higher-dimensional proximity constructions include relative-neighborhood graphs in $\mathbb{R}^3$ (Agarwal and Matoušek, 1992) and extensions of proximity regions to higher dimensions (Ceyhan, 2010). These graphs also have applications in pattern recognition, data analysis, and spatial-network modelling (Jaromczyk and Toussaint, 1992; Watanabe, 2008).

The stepping-stone graph was introduced by Kannangara, Tanin, Harwood and Karunasekera (2018) for movement analysis and subsequently developed in Kannangara, Tanin, Harwood and Karunasekera (2019). Those works study the parameterized family of planar proximity graphs obtained as the diversion parameter varies, its relationship with classical proximity graphs, and its application to movement-corridor analysis. We denote their parameter by α , reserving d for the ambient dimension.

In the original formulation, a planar point belongs to the diversion neighborhood when the sum of the α th powers of its distances to the two defining sites does not exceed the α th power of the distance between the sites. For a point set V , two points $p, q \in V$ are adjacent in the stepping-stone graph precisely when this neighborhood contains no point of $V \setminus \{p, q\}$. The diversion neighborhood reduces to the segment joining the sites at $\alpha = 1$ and equals the region associated with the Gabriel graph at $\alpha = 2$. It has been shown that the relative-neighborhood graph is the limiting member of the graph family and that the family is nested under edge inclusion (Kannangara et al., 2018, 2019). Although

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the original definition is planar, the same inequality defines a diversion neighborhood in every space $\mathbb{R}^d$ with dimension $d > 2$.

The remainder of the paper is organized as follows. Section 2 introduces the normalized diversion neighborhood and establishes its similarity factorization and elementary geometric properties in every dimension $d \geq 2$. Section 3 derives the boundary parametrization and a one-dimensional volume formula, recovers the degenerate and Gabriel graph cases, and identifies the relative-neighborhood limit. Section 4 proves strict monotonicity and derives the resulting graph inclusions and connectivity. Section 5 applies Devroye's regular-set framework to derive asymptotic expected-degree and graph-size results and examines planar Poisson edge-length distributions. Section 4 establishes strict region and volume monotonicity and derives the corresponding graph inclusions and connectivity results. Section 5 applies the probabilistic cases with Devroye's regular-sets results on the number of neighbors and Watanabe's approximation to the edge-length for a planar Poisson point processes, Section 6 provides a stable integral representation and independent numerical validation. Unless explicitly stated otherwise, the geometric and graph-theoretic results hold in every ambient dimension $d \geq 2$.

The principal contribution of this paper is a unified arbitrary-dimensional framework for the stepping-stone diversion neighborhood. For every $d \geq 2$, we derive its similarity factorization, obtain an explicit boundary parametrization and a one-dimensional integral for its normalized volume, and establish the resulting graph inclusions, connectivity properties, and probabilistic consequences. The volume is especially relevant under random-point models, where it determines empty-region probabilities and enters the constants governing expected graph size. To our knowledge, this collection of results has not previously been established for the stepping-stone family in arbitrary dimension.

The main difficulty is that the defining distance condition describes the region only implicitly. Direct numerical computation therefore treats every choice of dimension and parameter as a new multidimensional integration problem over a curved region. We instead place the two sites in a standard position and exploit the rotational symmetry around the line joining them. An explicit description of the resulting boundary profile then reduces the volume calculation to a one-dimensional integral.

The similarity reduction provides a normalization principle. Once the dimension d and the parameter α are fixed, changing the distance between the two sites changes only the scale of the diversion neighborhood, not its normalized shape, so its volume factorizes into the d th power of that distance and a constant depending only on α and d . The main theorem is the analytic reduction from the implicitly defined boundary to an explicit parametrization and then to a one-dimensional integral for this constant. We also derive a numerically stable form for its evaluation and recover the degenerate case, the Gabriel graph region case, and the relative-neighborhood limit.

We further prove, for every $d \geq 2$, that both the diversion neighborhoods and their volumes increase strictly with α , while the corresponding empty-region graphs decrease under edge inclusion. We establish strict containment in the Gabriel's graph neighborhood for $1 \leq \alpha < 2$, obtain dimension-dependent bounds for the normalized volume constants, and derive connectivity and classical graph inclusions. The numerical evaluation is independently validated through rejection-sampling experiments covering several dimensions and parameter values.

For $\alpha \geq 2$, the graph inclusions show in every ambient dimension that the defining empty-region condition need only be evaluated for Gabriel graph's edges that are not open-lune relative-neighborhood edges.

The relevance of the normalized volume constant is also reflected in probabilistic analyses of proximity graphs. Devroye's regular-set theorem shows that this single geometric constant determines the limiting expected degree for broad classes of sampling densities (Devroye, 1988). In the planar Poisson setting, the normalized area also determines Watanabe's graph-edge-length approximation (Watanabe, 2008). These consequences are derived in Section 5. The planar volume formula, the Delaunay-based construction, and Watanabe's edge-length approximation illustrate how the results obtained for general dimension specialize to established geometric, computational, and probabilistic results in the plane.

2. Definitions and geometric preliminaries

Throughout, $d \geq 2$ denotes the ambient dimension, and all points and regions are considered in $\mathbb{R}^d$. We write $\mathbf{0}$ for the origin, e_1 for the first standard basis vector, and

$$B(x, r) := \{z \in \mathbb{R}^d : \|z - x\| \leq r\} \quad (1)$$

for the closed Euclidean ball of radius r centered at $x \in \mathbb{R}^d$, and

$$B^\circ(x, r) := \{z \in \mathbb{R}^d : \|z - x\| < r\} \quad (2)$$

for the corresponding open ball. Furthermore, λ_d denotes d -dimensional Lebesgue measure, and

$$\kappa_d := \lambda_d(B(\mathbf{0}, 1)) = \frac{\pi^{d/2}}{\Gamma(1 + d/2)} \quad (3)$$

is the volume of the unit ball in $\mathbb{R}^d$.

Definition 1 (Stepping-stone diversion neighborhood). Let $p, q \in \mathbb{R}^d$ be distinct, and set

$$\ell_{p,q} := \|q - p\|, \quad u_{p,q} := \frac{q - p}{\ell_{p,q}}. \quad (4)$$

For $\alpha \geq 1$, the stepping-stone diversion neighborhood determined by p and q is

$$S_{SS,\alpha}(p, q) := \left\{ z \in \mathbb{R}^d : \|z - p\|^\alpha + \|z - q\|^\alpha \leq \ell_{p,q}^\alpha \right\}. \quad (5)$$

Its normalized region is

$$K_{SS,\alpha} := S_{SS,\alpha}(\mathbf{0}, e_1) = \left\{ z \in \mathbb{R}^d : \|z\|^\alpha + \|z - e_1\|^\alpha \leq 1 \right\}. \quad (6)$$

The normalized volume constant is

$$a_{d,SS}(\alpha) := \lambda_d(K_{SS,\alpha}). \quad (7)$$

For later comparison, the closed and open Gabriel regions determined by p and q are the closed and open balls with diameter pq , respectively:

$$G_{cl}(p, q) := B\left(\frac{p+q}{2}, \frac{\ell_{p,q}}{2}\right), \quad G_{op}(p, q) := B^\circ\left(\frac{p+q}{2}, \frac{\ell_{p,q}}{2}\right), \quad (8)$$

The closed and open relative-neighborhood lunes are, respectively,

$$L_{cl}(p, q) := B(p, \ell_{p,q}) \cap B(q, \ell_{p,q}), \quad L_{op}(p, q) := B^\circ(p, \ell_{p,q}) \cap B^\circ(q, \ell_{p,q}). \quad (9)$$

For a finite or locally finite set $V \subset \mathbb{R}^d$ and distinct $p, q \in V$, these graphs are defined by the property

$$pq \in E(GG(V)) \iff N(p, q) \cap (V \setminus \{p, q\}) = \emptyset, \quad (10)$$

where G denote the proximity graph and N its associated neighborhood.

Thus $GG_{cl}(V) \subseteq GG_{op}(V)$ and $RNG_{cl}(V) \subseteq RNG_{op}(V)$. Although the corresponding region boundaries have zero d -dimensional volume, the graph conventions may only differ when a third site lies on such a boundary.

Throughout the paper, $GG(V)$ denotes $GG_{cl}(V)$ and $RNG(V)$ denotes $RNG_{op}(V)$, consistently with the original stepping-stone construction and the empty-region conventions of Kannangara et al. (2018) and Cardinal et al. (2009). The alternative graphs $GG_{op}(V)$ and $RNG_{cl}(V)$ are always identified explicitly.

Definition 2 (Stepping-stone graph). For a finite or locally finite set $V \subset \mathbb{R}^d$, the stepping-stone graph with parameter α is the graph $G = (V, E)$ such that two distinct points $p, q \in V$ are adjacent if

$$S_{SS,\alpha}(p, q) \cap (V \setminus \{p, q\}) = \emptyset. \quad (11)$$

Proposition 1 (Similarity normalization and volume factorization). If Q is an orthogonal transformation satisfying $Qe_1 = u_{p,q}$, then

$$S_{SS,\alpha}(p, q) = p + \ell_{p,q} Q K_{SS,\alpha}. \quad (12)$$

Consequently,

$$\lambda_d(S_{SS,\alpha}(p, q)) = \ell_{p,q}^d a_{d,SS}(\alpha). \quad (13)$$

Proof. For $w \in \mathbb{R}^d$, substitute $z = p + \ell_{p,q} Qw$ in (5). The two normalized distances become $\|w\|$ and $\|w - e_1\|$, which proves (12). Translations and orthogonal transformations preserve Lebesgue measure, while dilation by $\ell_{p,q}$ multiplies d -dimensional volume by $\ell_{p,q}^d$, proving (13). $\square$

Thus it suffices to study the normalized region $K_{SS,\alpha}$. Figure 1 shows the same three representative parameter values in two and three dimensions. The dashed planar boundary is the limiting relative-neighborhood lune.

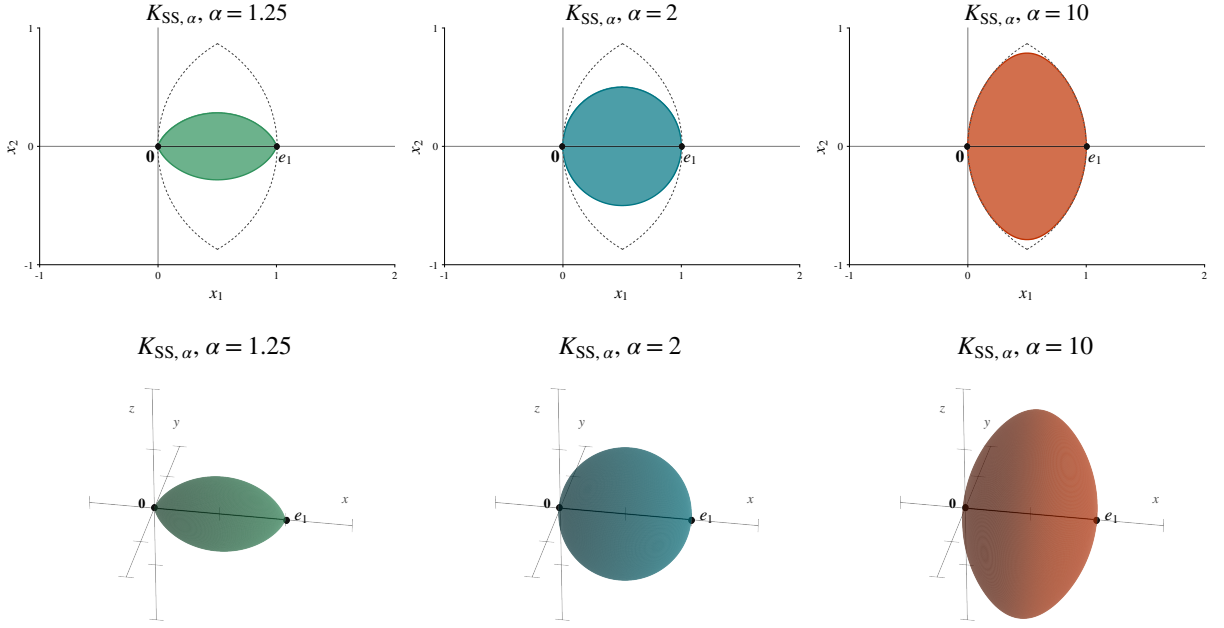


Figure 1: Normalized stepping-stone diversion neighborhoods for $\alpha = 1.25, 2$, and 10 . Top: planar regions, with the limiting relative-neighborhood lune shown by the dashed boundary. Bottom: three-dimensional regions for the same parameter values. The defining sites are fixed at $\mathbf{0}$ and e_1 .

To complement the static views in Figure 1, the computational companion (Espino-Montelongo and Maravillo, 2026) provides an interactive three-dimensional visualization of the normalized diversion neighborhood.

Proposition 2 (Elementary geometry). *For every $\alpha \geq 1$, the region $S_{SS,\alpha}(p, q)$ is compact and convex. It is centrally symmetric about*

$$m := \frac{p + q}{2}, \quad (14)$$

rotationally symmetric about the line through p and q and invariant under every Euclidean isometry fixing the line through p and q pointwise. If $\alpha > 1$, then $S_{SS,\alpha}(p, q)$ has nonempty interior and is strictly convex.

Proof. The function

$$F_{p,q,\alpha}(z) := \|z - p\|^\alpha + \|z - q\|^\alpha \quad (15)$$

is continuous and convex for $\alpha \geq 1$. Hence its sublevel set $S_{SS,\alpha}(p, q)$ is closed and convex. Its defining inequality implies

$$\|z - p\| \leq \ell_{p,q}, \quad \|z - q\| \leq \ell_{p,q}, \quad (16)$$

so the region is bounded and therefore compact.

The reflection $z \mapsto p + q - z$ exchanges the two distance terms and preserves the region, invariant under every Euclidean isometry fixing the line through p and q pointwise and therefore proving central symmetry. Every rotation about the line through p and q preserves both distances and therefore preserves the region.

For $\alpha > 1$, the map $z \mapsto \|z - p\|^\alpha$ is strictly convex, and so is $F_{p,q,\alpha}$. If $z_1 \neq z_2$ belong to the region and $t \in (0, 1)$, then

$$F_{p,q,\alpha}(tz_1 + (1-t)z_2) < tF_{p,q,\alpha}(z_1) + (1-t)F_{p,q,\alpha}(z_2) \leq \ell_{p,q}^\alpha. \quad (17)$$

Thus the open segment joining z_1 and z_2 lies in the interior. Moreover,

$$F_{p,q,\alpha}(m) = 2 \left(\frac{\ell_{p,q}}{2} \right)^\alpha = 2^{1-\alpha} \ell_{p,q}^\alpha < \ell_{p,q}^\alpha, \quad (18)$$

so the interior is nonempty. $\square$

3. Boundary parametrization and volume

All results in this section are valid for every ambient dimension $d \geq 2$; the planar formula is recorded afterward only as a special case.

3.1. Boundary parametrization

We work in the normalized configuration $(p, q) = (\mathbf{0}, e_1)$ and write

$$z = (X, Z), \quad X \in \mathbb{R}, \quad Z \in \mathbb{R}^{d-1}, \quad (19)$$

with transverse radius $Y := \|Z\|$. The boundary is specified implicitly by

$$(X^2 + Y^2)^{\alpha/2} + ((X-1)^2 + Y^2)^{\alpha/2} = 1. \quad (20)$$

The following result turns this implicit equation into the boundary profile used throughout the volume calculation.

Proposition 3 (Boundary parametrization). *Let $\alpha > 1$ and set $u_0 := 2^{-1/\alpha}$. For $u \in [u_0, 1]$, define*

$$\rho(u) := (1 - u^\alpha)^{1/\alpha}, \quad X_\alpha(u) := \frac{1 + u^2 - \rho(u)^2}{2}, \quad (21)$$

and define the midpoint-centered axial coordinate and transverse radius by

$$x_\alpha(u) := X_\alpha(u) - \frac{1}{2} = \frac{u^2 - (1 - u^\alpha)^{2/\alpha}}{2}, \quad (22)$$

$$y_\alpha(u) := \sqrt{u^2 - X_\alpha(u)^2} = \frac{1}{2} \sqrt{4u^2 - (1 + u^2 - (1 - u^\alpha)^{2/\alpha})^2}. \quad (23)$$

Then $(X_\alpha(u), y_\alpha(u))$ traces the complete boundary profile in the half-space $X \geq 1/2$; orthogonal transformations fixing the e_1 -axis give the complete right boundary of $K_{SS,\alpha}$. Moreover,

$$x_\alpha(u_0) = 0, \quad x_\alpha(1) = \frac{1}{2}, \quad (24)$$

and

$$x'_\alpha(u) = u + u^{\alpha-1}(1 - u^\alpha)^{(2-\alpha)/\alpha} > 0 \quad (u_0 \leq u < 1). \quad (25)$$

Thus x_α maps $[u_0, 1]$ continuously and strictly increasingly onto $[0, 1/2]$.

Proof. Parametrize a boundary point by its distance to the first site, $u := \|z\|$. Its distance to the second site is then $\rho(u)$ because $u^\alpha + \rho(u)^\alpha = 1$. On the right half, $u \geq \rho(u)$, which is equivalent to $u \geq u_0$; the endpoint $u = 1$ is the site e_1 . The two sphere equations

$$X^2 + Y^2 = u^2, \quad (X-1)^2 + Y^2 = \rho(u)^2 \quad (26)$$

give $2X - 1 = u^2 - \rho(u)^2$, and hence (22)–(23). Conversely, for every $u \in [u_0, 1]$ we have

$$u + \rho(u) \geq u^\alpha + \rho(u)^\alpha = 1, \quad |u - \rho(u)| \leq 1, \quad (27)$$

so the two spheres intersect and (23) is real. These equations therefore construct a boundary point, and every boundary point in the right half has exactly one value of u . The endpoint identities follow by substitution. Differentiating (22) gives (25), whose two terms are positive on $[u_0, 1)$. For $\alpha > 2$, the second term has an endpoint singularity at $u = 1$. Since $1 - u^\alpha \sim \alpha(1 - u)$ and $2/\alpha - 1 > -1$, the singularity is integrable; the integrals below are interpreted as improper integrals in that case. $\square$

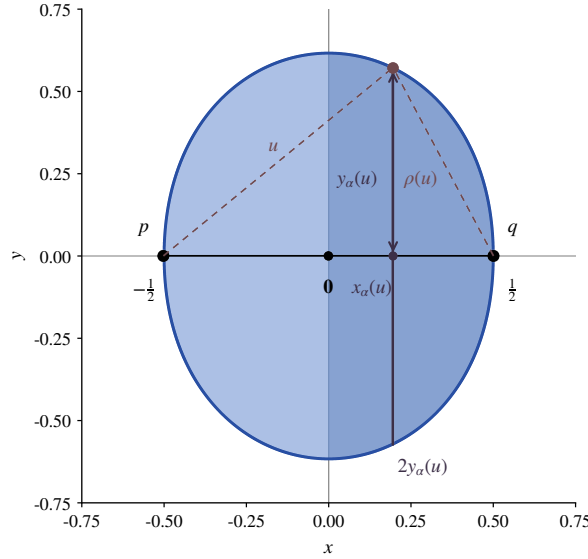


Figure 2: Midpoint-centered slicing of $K_{SS,\alpha}$. The right half is parametrized by $x = x_\alpha(u)$, and the transverse radius is $y_\alpha(u)$.

3.2. One-dimensional volume formula

Theorem 1 (One-dimensional volume formula). *Let $d \geq 2$ and $\alpha > 1$. Then*

$$a_{d,SS}(\alpha) = 2\kappa_{d-1} \int_{2^{-1/\alpha}}^1 y_\alpha(u)^{d-1} x'_\alpha(u) du, \quad (28)$$

where y_α and x'_α are given by (23) and (25). Together with Proposition 1, this determines the volume for every pair of distinct sites in $\mathbb{R}^d$.

Proof. Every point $z = (X, Z)$ in the normalized region satisfies $0 \leq X \leq 1$. Indeed, (6) implies

$$\|z\| \leq 1 \quad \text{and} \quad \|z - e_1\| \leq 1, \quad (29)$$

since both terms on its left-hand side are nonnegative. If $X < 0$, then

$$\|z - e_1\| \geq |1 - X| = 1 - X > 1, \quad (30)$$

whereas, if $X > 1$, then

$$\|z\| \geq |X| = X > 1. \quad (31)$$

Both cases contradict the preceding necessary inequalities. Hence the normalized region is contained in the slab $0 \leq X \leq 1$.

For $x \in [0, 1/2]$, define the transverse section

$$S_x := \{Z \in \mathbb{R}^{d-1} : (x + 1/2, Z) \in K_{SS,\alpha}\}. \quad (32)$$

By Proposition 2, S_x is compact, convex, and invariant under every orthogonal transformation of $\mathbb{R}^{d-1}$. Moreover, $0 \in S_x$, since the segment $[0, e_1]$ is contained in the normalized region.

Let

$$R(x) := \max \{\|Z\| : Z \in S_x\}, \quad (33)$$

where the maximum exists by compactness. If $Z_* \in S_x$ satisfies $\|Z_*\| = R(x)$, orthogonal invariance implies that

$$\{Z \in \mathbb{R}^{d-1} : \|Z\| = R(x)\} \subseteq S_x. \quad (34)$$

Since S_x is convex, it contains the convex hull of this sphere, namely the closed ball of radius $R(x)$. Conversely, the definition of $R(x)$ implies that every $Z \in S_x$ satisfies $\|Z\| \leq R(x)$. Hence

$$S_x = \{Z \in \mathbb{R}^{d-1} : \|Z\| \leq R(x)\}, \quad (35)$$

and therefore

$$\lambda_{d-1}(S_x) = \kappa_{d-1} R(x)^{d-1}. \quad (36)$$

By Proposition 3, the boundary of the right half is given by $x = x_\alpha(u)$ and $R(x_\alpha(u)) = y_\alpha(u)$ for $u \in [u_0, 1]$, and x_α is a valid change of variables from $[u_0, 1]$ onto $[0, 1/2]$.

By Cavalieri's principle,

$$\lambda_d(K_{SS,\alpha} \cap \{X \geq 1/2\}) = \int_0^{1/2} \kappa_{d-1} R(x)^{d-1} dx. \quad (37)$$

Changing variables from x to u gives

$$\lambda_d(K_{SS,\alpha} \cap \{X \geq 1/2\}) = \kappa_{d-1} \int_{u_0}^1 y_\alpha(u)^{d-1} x'_\alpha(u) du. \quad (38)$$

Finally, the reflection $(X, Z) \mapsto (1 - X, Z)$ preserves $K_{SS,\alpha}$ and exchanges the two halves. The midpoint hyperplane has zero d -dimensional Lebesgue measure, so multiplying the last display by 2 proves (28). $\square$

Remark 1 (Planar specialization). For $d = 2$, $\kappa_{d-1} = \kappa_1 = 2$, and therefore

$$a_{2,SS}(\alpha) = 4 \int_{2^{-1/\alpha}}^1 y_\alpha(u) x'_\alpha(u) du. \quad (39)$$

3.3. Special and limiting cases

Proposition 4 (Degenerate and Gabriel graph region cases). *For every $d \geq 2$,*

$$a_{d,SS}(1) = 0, \quad (40)$$

whereas

$$a_{d,SS}(2) = \frac{\kappa_d}{2^d}. \quad (41)$$

At $\alpha = 1$, the diversion neighborhood is the segment $[p, q]$; at $\alpha = 2$, it is the closed ball with diameter pq . In particular, $SSG_2(V) = GG(V)$ for every finite or locally finite point set V . At $\alpha = 1$, two sites are adjacent exactly when the open segment joining them contains no other point of V ; consequently, $SSG_1(V)$ is complete whenever no three points of V are collinear.

Proof. At $\alpha = 1$, the triangle inequality shows that

$$\|z - p\| + \|z - q\| \leq \ell_{p,q} \quad (42)$$

holds precisely on the segment joining p and q , which has zero d -dimensional measure. At $\alpha = 2$, let $m = (p+q)/2$. The parallelogram identity gives

$$\|z - p\|^2 + \|z - q\|^2 = 2\|z - m\|^2 + 2\left(\frac{\ell_{p,q}}{2}\right)^2. \quad (43)$$

Thus the region is the ball centered at m with radius $\ell_{p,q}/2$, and its normalized volume is $\kappa_d/2^d$. The graph identity follows from the closed-ball convention in the definition of $GG(V)$. The characterization at $\alpha = 1$ follows by deleting the two defining sites from the segment $[p, q]$. $\square$

Proposition 5 (Convergence to the relative-neighborhood lune). *Define the normalized relative-neighborhood volume constant by*

$$a_{d,\text{RNG}} := \lambda_d(L_{\text{op}}(\mathbf{0}, e_1)) = \lambda_d(L_{\text{cl}}(\mathbf{0}, e_1)). \quad (44)$$

As $\alpha \rightarrow \infty$, the diversion neighborhoods converge from within to the open relative-neighborhood lune in the precise sense that

$$\bigcup_{1 < \alpha < \infty} (S_{\text{SS},\alpha}(p, q) \setminus \{p, q\}) = L_{\text{op}}(p, q). \quad (45)$$

Moreover,

$$\lim_{\alpha \rightarrow \infty} a_{d,\text{SS}}(\alpha) = a_{d,\text{RNG}} = \kappa_d I_{3/4} \left(\frac{d+1}{2}, \frac{1}{2} \right). \quad (46)$$

where $I_{3/4}$ is the regularized incomplete beta function.

Proof. For $z \notin \{p, q\}$, the defining inequality implies the strict bounds

$$\|z - p\| < \ell_{p,q}, \quad \|z - q\| < \ell_{p,q}. \quad (47)$$

Thus every finite-parameter diversion neighborhood, after removing its defining sites, is contained in $L_{\text{op}}(p, q)$.

Conversely, let $z \in L_{\text{op}}(p, q)$. Then

$$0 \leq \frac{\|z - p\|}{\ell_{p,q}} < 1, \quad 0 \leq \frac{\|z - q\|}{\ell_{p,q}} < 1, \quad (48)$$

and consequently

$$\left(\frac{\|z - p\|}{\ell_{p,q}} \right)^\alpha + \left(\frac{\|z - q\|}{\ell_{p,q}} \right)^\alpha \rightarrow 0 \quad \text{as } \alpha \rightarrow \infty. \quad (49)$$

Hence, for all sufficiently large α ,

$$\|z - p\|^\alpha + \|z - q\|^\alpha \leq \ell_{p,q}^\alpha, \quad (50)$$

so $z \in S_{\text{SS},\alpha}(p, q)$.

This proves (45). Since membership is eventually constant at every point, the neighborhoods lie in the bounded closed lune, and deleting their two defining sites does not change Lebesgue measure, dominated convergence yields

$$\lim_{\alpha \rightarrow \infty} \lambda_d(S_{\text{SS},\alpha}(p, q)) = \lambda_d(L_{\text{cl}}(p, q)). \quad (51)$$

The volume of the unit relative-neighborhood lune in arbitrary dimension was previously represented as a radial integral by Devroye (1988, Sec. 5, p. 59). Equivalently, in normalized coordinates, the midpoint hyperplane partitions $L_{\text{cl}}(\mathbf{0}, e_1)$ into two congruent unit-ball caps of height $1/2$. The standard hyperspherical-cap formula of Li (2011) gives one cap volume

$$\frac{\kappa_d}{2} I_{3/4} \left(\frac{d+1}{2}, \frac{1}{2} \right). \quad (52)$$

Doubling proves (46). For example, in dimension three the benchmarks corresponding to the Gabriel graph and relative-neighborhood-graph benchmarks are $a_{3,\text{SS}}(2) = \pi/6$ and $a_{3,\text{RNG}} = 5\pi/12$, respectively. $\square$

4. Monotonicity and graph consequences

The region, volume, and graph results in this section hold in arbitrary dimension. The Delaunay containment in the final remark is also dimension independent; only the linear candidate bound and the cited stepping-stone algorithm are specific to the plane.

4.1. Strict region and volume monotonicity

Theorem 2 (Strict monotonicity). *For every $1 \leq \alpha_1 < \alpha_2$ and every distinct $p, q \in \mathbb{R}^d$,*

$$S_{SS,\alpha_1}(p, q) \subsetneq S_{SS,\alpha_2}(p, q), \quad (53)$$

and the set difference has positive d -dimensional measure. Consequently,

$$a_{d,SS}(\alpha_1) < a_{d,SS}(\alpha_2). \quad (54)$$

Proof. For $a, b \geq 0$, the map

$$r \mapsto (a^r + b^r)^{1/r}, \quad r > 0, \quad (55)$$

is nonincreasing. If $z \in S_{SS,\alpha_1}(p, q)$, then

$$(\|z - p\|^{\alpha_1} + \|z - q\|^{\alpha_1})^{1/\alpha_1} \leq \ell_{p,q}. \quad (56)$$

The monotonicity of this map gives

$$(\|z - p\|^{\alpha_2} + \|z - q\|^{\alpha_2})^{1/\alpha_2} \leq (\|z - p\|^{\alpha_1} + \|z - q\|^{\alpha_1})^{1/\alpha_1} \leq \ell_{p,q}, \quad (57)$$

so $z \in S_{SS,\alpha_2}(p, q)$. Taking Lebesgue measure and using (13) gives the volume monotonicity.

To prove strictness, let $m = (p+q)/2$ and choose a unit vector v perpendicular to $q - p$. On the midpoint hyperplane, a point $z = m + tv$ belongs to $S_{SS,\alpha}(p, q)$ precisely when

$$|t| \leq r_\alpha, \quad r_\alpha := \ell_{p,q} \sqrt{2^{-2/\alpha} - \frac{1}{4}}. \quad (58)$$

The function $\alpha \mapsto 2^{-2/\alpha}$ is strictly increasing, so $r_{\alpha_1} < r_{\alpha_2}$. Choose t strictly between these two radii. Then $m + tv$ lies in the interior of the larger region and outside the closed smaller region. A neighborhood of this point is therefore contained in the set difference, which has positive d -dimensional measure. The strict inequality for the normalized volumes follows from (13). $\square$

Corollary 1 (Continuous parameter calibration). *For every $d \geq 2$, the map $\alpha \mapsto a_{d,SS}(\alpha)$ is continuous and strictly increasing on $[1, \infty)$, with image $[0, a_{d,RNG})$. Consequently, every target volume $c \in (0, a_{d,RNG})$ determines a unique finite parameter $\alpha > 1$.*

Proof. Let $\alpha_n \rightarrow \alpha \geq 1$. Away from the boundary of $K_{SS,\alpha}$, continuity of the defining inequality implies eventual agreement of membership in K_{SS,α_n} and $K_{SS,\alpha}$. This boundary has Lebesgue measure zero: for $\alpha > 1$ it is the boundary of a convex body, while for $\alpha = 1$ the entire region is a line segment. All these regions are contained in the bounded closed relative-neighborhood lune, so dominated convergence proves continuity of their volumes. The image and uniqueness now follow from Propositions 4 and 5 and Theorem 2. $\square$

Corollary 2 (Bounds and Gabriel graph region comparison). *For every $d \geq 2$, the following holds:*

$$\begin{cases} a_{d,SS}(\alpha) = 0, & \alpha = 1, \\ 0 < a_{d,SS}(\alpha) < \kappa_d/2^d, & 1 < \alpha < 2, \\ a_{d,SS}(\alpha) = \kappa_d/2^d, & \alpha = 2, \\ \kappa_d/2^d < a_{d,SS}(\alpha) < a_{d,RNG}, & 2 < \alpha < \infty. \end{cases} \quad (59)$$

Moreover, for every distinct $p, q \in \mathbb{R}^d$,

$$S_{SS,\alpha}(p, q) \subsetneq GG_{cl}(p, q) \quad (1 \leq \alpha < 2), \quad (60)$$

whereas $GG_{cl}(p, q) \subsetneq S_{SS,\alpha}(p, q)$ for every $2 < \alpha < \infty$.

Proof. Combine Propositions 4 and 5 and Theorem 2. The strict upper bound by $a_{d,RNG}$ at every finite parameter follows because a strictly increasing function cannot attain its limiting value at a finite argument. $\square$

4.2. Monotonicity and graph inclusions

For $i \in \{1, 2\}$, let $S_i(p, q)$ be a region assigned to each pair of distinct sites, and let $G_i(V) = (V, E_i(V))$ join p and q precisely when $S_i(p, q)$ contains no point of $V \setminus \{p, q\}$. Directly from this definition, whenever the region inclusion below holds for every distinct pair, so does the edge inclusion:

$$S_1(p, q) \setminus \{p, q\} \subseteq S_2(p, q) \setminus \{p, q\} \implies E_2(V) \subseteq E_1(V). \quad (61)$$

Indeed, if the intersection of $S_2(p, q)$ with $V \setminus \{p, q\}$ is empty, then the corresponding intersection with $S_1(p, q)$ is also empty.

Let $\text{MST}(V)$ denote a Euclidean minimum spanning tree of a nonempty finite point set V , and let $\text{Del}(V)$ denote its Delaunay graph (Mitchell and Mulzer, 2017). With the closed Gabriel-ball and open relative-neighborhood-lune conventions adopted above, the standard inclusions are

$$\text{MST}(V) \subseteq \text{RNG}(V) \subseteq \text{GG}(V) \subseteq \text{Del}(V). \quad (62)$$

The hierarchy is reviewed by Jaromczyk and Toussaint (1992); the closed-ball Gabriel convention and its Delaunay inclusion are explicit in Matula and Sokal (1980) and Cardinal et al. (2009). In particular, an empty closed diametral ball itself certifies the Delaunay empty-ball condition, including configurations with cospherical sites. The stepping-stone family fits into this chain as follows.

The stepping-stone family fits into this chain as follows.

Proposition 6 (Monotonicity and classical proximity-graph inclusions). *If $1 \leq \alpha_1 < \alpha_2$, then*

$$E_{\alpha_2}(V) \subseteq E_{\alpha_1}(V). \quad (63)$$

Thus $\{\text{SSG}_\alpha(V) : \alpha \geq 1\}$ is a monotonically decreasing family of graphs under edge inclusion, although the inclusion need not be strict for a fixed point set. Furthermore, for every nonempty finite $V \subset \mathbb{R}^d$ and every finite $\alpha \geq 1$,

$$\text{MST}(V) \subseteq \text{RNG}(V) \subseteq \text{SSG}_\alpha(V). \quad (64)$$

If $1 \leq \alpha \leq 2$, then

$$\text{MST}(V) \subseteq \text{RNG}(V) \subseteq \text{GG}(V) \subseteq \text{SSG}_\alpha(V). \quad (65)$$

If $\alpha \geq 2$, then

$$\text{MST}(V) \subseteq \text{RNG}(V) \subseteq \text{SSG}_\alpha(V) \subseteq \text{GG}(V) \subseteq \text{Del}(V). \quad (66)$$

Proof. The first edge inclusion follows directly from Theorem 2 and the preceding implication; it is the region-inclusion formulation of the nestedness proved for the original d -spectrum by Kannangara et al. (2018, 2019).

For $z \notin \{p, q\}$, the inequality defining $S_{\text{SS},\alpha}(p, q)$ implies

$$\|z - p\| < \ell_{p,q}, \quad \|z - q\| < \ell_{p,q}. \quad (67)$$

Thus $S_{\text{SS},\alpha}(p, q) \setminus \{p, q\} \subset L_{\text{op}}(p, q)$, and the same implication gives $\text{RNG}(V) \subseteq \text{SSG}_\alpha(V)$. The remaining stepping-stone inclusions follow from Proposition 4 and Theorem 2; the other inclusions are classical. $\square$

Corollary 3 (Connectivity in arbitrary dimension). *For every nonempty finite $V \subset \mathbb{R}^d$ and every finite $\alpha \geq 1$, the graph $\text{SSG}_\alpha(V)$ is connected.*

Proof. Connectivity follows from

$$\text{MST}(V) \subseteq \text{RNG}_{\text{op}}(V) \subseteq \text{SSG}_\alpha(V). \quad (68)$$

$\square$

Corollary 4 (Finite-parameter stabilization). *For every finite $V \subset \mathbb{R}^d$, there exists a finite parameter $\alpha_*(V) \geq 1$ such that*

$$\text{SSG}_\alpha(V) = \text{RNG}_{\text{op}}(V) \quad \text{for all } \alpha \geq \alpha_*(V). \quad (69)$$

Proof. For every pair $p, q \in V$ that is not an edge of $\text{RNG}_{\text{op}}(V)$, choose a point $z \in L_{\text{op}}(p, q) \cap (V \setminus \{p, q\})$. By Proposition 5, that point belongs to $S_{\text{SS},\alpha}(p, q)$ for every sufficiently large α . There are only finitely many such pairs, so their individual thresholds have a finite maximum $\alpha_*(V)$; if there are no such pairs, take $\alpha_*(V) = 1$. Beyond this threshold, no non-relative-neighborhood edge remains, while Proposition 6 preserves every relative-neighborhood edge. $\square$

Remark 2 (Dependence and provenance of the stabilization threshold). In the planar case, finite stabilization is already implicit in the finite deletion thresholds of the d -spectrum introduced by Kannangara et al. (2018, Def. 3.12); the preceding argument extends the observation to every fixed ambient dimension. The threshold cannot be chosen uniformly over finite point sets.

Remark 3 (Planar Delaunay starting graph). For $V \subset \mathbb{R}^2$ and $\alpha \geq 2$, $\text{SSG}_\alpha(V)$ is a subgraph of the Gabriel graph and hence of the Delaunay graph. Therefore, if V is in general position and $n = |V| \geq 3$, the defining empty-region condition need only be tested for at most $3n - 6$ Delaunay edges, rather than for all $\binom{n}{2}$ pairs. Testing each candidate against the remaining $n - 2$ sites still gives a naive $O(n^2)$ procedure. In their planar d -spectrum algorithm, Kannangara et al. (2018, 2019) use the Delaunay triangulation as the starting graph. Although the Delaunay containment remains valid in higher dimensions, it does not yield a subquadratic worst-case candidate set, since Delaunay graphs can already have $\Theta(n^2)$ edges in $\mathbb{R}^3$ (Agarwal and Matoušek, 1992). The obstruction persists even after restricting to **Gabriel candidates** since Gabriel graphs themselves can have $\Theta(n^2)$ edges in $\mathbb{R}^3$ (Chazelle, Edelsbrunner, Guibas, Hershberger, Seidel and Sharir, 1994, Lemma 5.1). Hence the candidate-edge characterization is not, by itself, a subquadratic construction algorithm in higher dimensions.

5. Probabilistic consequences

The normalized volume is also the parameter that enters the expected-size theory of regular empty-region graphs. The first part of this section records the arbitrary-dimensional consequences of Devroye's theorem. The second part is an explicitly planar specialization of the graph-edge-length approximation of Watanabe (2008). The verification below uses compactness, midpoint symmetry, and similarity equivariance from Section 2; convexity itself is not a hypothesis in Devroye's definition.

5.1. Expected size in arbitrary dimension

Proposition 7 (Regularity in the sense of Devroye). *For every fixed $\alpha > 1$, the stepping-stone construction defines a regular set in the sense of Devroye (1988), with normalized volume $a_{d,\text{SS}}(\alpha)$.*

Proof. In Devroye's terminology, take the bounded target set

$$T_\alpha := \{(u, v) \in \mathbb{R} \times [0, \infty) : (u^2 + v^2)^{\alpha/2} + ((1 - u)^2 + v^2)^{\alpha/2} \leq 1\}. \quad (70)$$

By Proposition 2, it is bounded and symmetric about $u = 1/2$. Given distinct $p, q \in \mathbb{R}^d$, the translation, rotation, and rescaling in Proposition 1 reduce membership of any point $z \in \mathbb{R}^d$ in $S_{\text{SS},\alpha}(p, q)$ to membership of the pair formed by its axial coordinate and its distance from that axis in T_α . Rotating T_α about the first coordinate axis produces $K_{\text{SS},\alpha}$, whose volume is $a_{d,\text{SS}}(\alpha)$. These are precisely the boundedness, symmetry, and similarity requirements in Devroye's definition of a regular set. $\square$

Corollary 5 (Expected number of neighbors). *Let $X_1, \dots, X_n$ be independent points with a common density on $\mathbb{R}^d$, and let $N_n(X_1)$ denote the number of stepping-stone neighbors of X_1 for a fixed $\alpha > 1$. Then, for almost every x ,*

$$\lim_{n \rightarrow \infty} \mathbb{E}[N_n(X_1) \mid X_1 = x] = \frac{\kappa_d}{a_{d,\text{SS}}(\alpha)}. \quad (71)$$

If M_n denotes the total number of undirected edges, then

$$\liminf_{n \rightarrow \infty} \frac{\mathbb{E}[M_n]}{n} \geq \frac{\kappa_d}{2a_{d,\text{SS}}(\alpha)}. \quad (72)$$

Proof. By Proposition 7, the volume in Devroye's regular-set theorem is $a_{d,SS}(\alpha)$, while the volume denoted there by V_d is κ_d . Substitution in the conditional degree conclusion of that theorem gives (71). Its edge-count conclusion gives (72); the factor $1/2$ appears because the sum of the vertex degrees counts every undirected edge twice. $\square$

5.2. Planar Poisson specialization

We now specialize to $d = 2$ and use the homogeneous Poisson model of Watanabe (2008), with point intensity $\rho > 0$. For brevity set

$$a := a_{2,SS}(\alpha), \quad \alpha > 1. \quad (73)$$

Proposition 8 (Planar edge-length approximation). *Within Watanabe's approximation, let L denote the modeled edge length for the planar stepping-stone graph generated by a homogeneous Poisson process of intensity ρ . Its distribution function is*

$$F_L(r) = 1 - \exp\{-\rho ar^2\}, \quad r \geq 0, \quad (74)$$

and its density is

$$f_L(r) = 2\rho ar \exp\{-\rho ar^2\}, \quad r > 0. \quad (75)$$

Consequently

$$\mathbb{E}[L] = \frac{\sqrt{\pi}}{2\sqrt{\rho a}}, \quad \text{Var}(L) = \frac{4 - \pi}{4\rho a}. \quad (76)$$

Proof. For a pair of sites at distance r , the area factorization (13) gives the area of the diversion neighborhood

$$A(r) = ar^2. \quad (77)$$

The Poisson probability that this region contains no point is

$$\mathbb{P}\{0 \text{ points in the search region of area } A(r)\} = \exp\{-\rho A(r)\} = \exp\{-\rho ar^2\}. \quad (78)$$

Within Watanabe's approximation, the probability that the restricted search has found at least one point by distance r is the complement of this void probability; this gives (74). Differentiation with respect to r gives (75).

For completeness, the same gamma-integral calculation gives

$$\mathbb{E}[L] = 2\rho a \int_0^\infty r^2 e^{-\rho ar^2} dr = \frac{\Gamma(3/2)}{\sqrt{\rho a}} = \frac{\sqrt{\pi}}{2\sqrt{\rho a}}. \quad (79)$$

Moreover,

$$\mathbb{E}[L^2] = 2\rho a \int_0^\infty r^3 e^{-\rho ar^2} dr = \frac{1}{\rho a}, \quad (80)$$

so subtracting $\mathbb{E}[L]^2$ yields the variance in (76). At $\alpha = 2$,

$$a_{2,SS}(2) = \pi/4, \quad (81)$$

and the formula recovers Watanabe's formula for the Gabriel graph. As $\alpha \rightarrow \infty$,

$$a_{2,SS}(\alpha) \longrightarrow \frac{2\pi}{3} - \frac{\sqrt{3}}{2}, \quad (82)$$

which recovers his approximation for the relative-neighborhood graph. $\square$

6. Numerical evaluation and validation

6.1. Stable quadrature

The formula in Theorem 1 is suitable for numerical evaluation, although its original form is not always the most stable one. For $\alpha > 2$, the derivative in (25) diverges as $u \rightarrow 1$, but the transverse radius simultaneously tends to zero. A direct endpoint expansion gives that

$$y_\alpha(u)^{d-1} x'_\alpha(u) \text{ is of order } (1-u)^{(d+1)/\alpha-1}. \quad (83)$$

Therefore, the complete integrand is unbounded only when $\alpha > d+1$, and it remains integrable for every finite $\alpha > 1$. Nevertheless, adaptive quadrature may lose accuracy near the endpoint, which motivates the following reparametrization.

Proposition 9 (Stable quadrature). *For $\alpha > 1$, define*

$$u(t) := (1 - t^\alpha)^{1/\alpha}, \quad (84)$$

and

$$\tilde{y}_\alpha(t) := \frac{1}{2} \sqrt{(u(t) + t - 1)(u(t) + t + 1)(1 + u(t) - t)(1 - u(t) + t)}. \quad (85)$$

Then

$$a_{d,SS}(\alpha) = 2\kappa_{d-1} \int_0^{2^{-1/\alpha}} \tilde{y}_\alpha(t)^{d-1} (t + u(t)^{2-\alpha} t^{\alpha-1}) dt, \quad (86)$$

and the integrand has no endpoint singularity.

Proof. Parametrize the right boundary by the distance to the second site,

$$t = (1 - u^\alpha)^{1/\alpha}, \quad u = (1 - t^\alpha)^{1/\alpha}. \quad (87)$$

Then $0 \leq t \leq 2^{-1/\alpha}$ and $x = (u^2 - t^2)/2$. Since $u'(t) = -u(t)^{1-\alpha} t^{\alpha-1}$,

$$\frac{dx}{dt} = -(t + u^{2-\alpha} t^{\alpha-1}). \quad (88)$$

The coordinate $x(t)$ decreases from $1/2$ to 0 , so the slicing formula uses the positive element $-dx = |dx/dt| dt$. Substitution in (28) gives (86). At $t = 0$, the second term in the Jacobian tends to zero for $\alpha > 1$, which removes the endpoint singularity. The factorized expression for $\tilde{y}_\alpha(t)$ follows from

$$4u^2 - (1 + u^2 - t^2)^2 = (u + t - 1)(u + t + 1)(1 + u - t)(1 - u + t). \quad (89)$$

□

Evaluating the transverse radius in this factored form reduces the cancellation caused by subtracting two nearly equal squared quantities; all four factors are nonnegative on the integration interval. The degenerate case is evaluated separately using $a_{d,SS}(1) = 0$ from Proposition 4.

6.2. Numerical curves and benchmark values

Figure 3 plots the volume constants themselves. The curves agree with the special and limiting values proved above: they start at zero when $\alpha = 1$, pass through the dimension-specific Gabriel graph value at $\alpha = 2$, increase strictly, and approach $a_{d,RNG}$ as $\alpha \rightarrow \infty$. Numerically, the two low-dimensional crossings shown in the figure satisfy $a_{2,SS}(\alpha) = a_{3,SS}(\alpha)$ at $\alpha \approx 7.118390847$ and $a_{2,SS}(\alpha) = a_{4,SS}(\alpha)$ at $\alpha \approx 23.643510648$. These intersections are numerical observations.

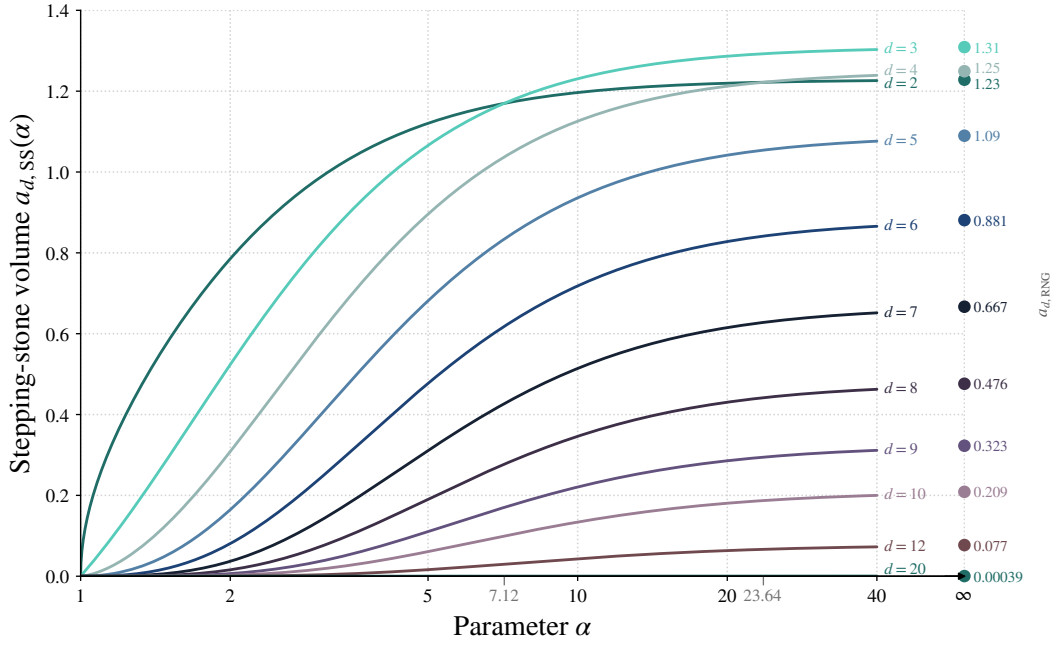


Figure 3: Numerical values of $a_{d,ss}(\alpha)$ for several dimensions on a logarithmic α -axis. The right endpoint shows $a_{d,RNG}$, the relative-neighborhood limit. The auxiliary labels 7.12 and 23.64 mark the two numerical intersections described in the text.

6.3. Independent rejection-sampling validation

The Monte Carlo calculation evaluates the original defining inequality in (6); it does not use the boundary parametrization or either quadrature formula.

The region is contained in the cylinder

$$C_{d,\alpha} := [0, 1] \times B_{d-1}(0, r_\alpha), \quad r_\alpha := \sqrt{2^{-2/\alpha} - \frac{1}{4}}. \quad (90)$$

Here $B_{d-1}(0, r_\alpha)$ denotes the closed ball of radius r_α in $\mathbb{R}^{d-1}$. Indeed, the defining inequality gives $0 \leq X \leq 1$. If (X, Z) belongs to the normalized region, midpoint symmetry in Proposition 2 shows that its reflection $(1 - X, Z)$ also belongs to it; convexity then places their midpoint $(1/2, Z)$ in the region. The largest possible transverse norm is therefore attained in the midpoint slice, where the defining equality gives $\|Z\| = r_\alpha$. Consequently,

$$V_{d,\alpha} := \lambda_d(C_{d,\alpha}) = \kappa_{d-1} r_\alpha^{d-1}. \quad (91)$$

If H of N independent uniform points in this cylinder satisfy (6), then

$$\hat{a}^{\text{MC}} = V_{d,\alpha} \frac{H}{N}, \quad \hat{\text{se}} = V_{d,\alpha} \sqrt{\frac{\hat{p}(1 - \hat{p})}{N}}, \quad \hat{p} := \frac{H}{N}. \quad (92)$$

Thus the estimator is unbiased, and its reported standard error is the usual binomial plug-in estimate.

We evaluated 18 parameter combinations with $N = 10^8$ samples each, for a total of 1.8×10^9 membership tests. The parameters were $\alpha = 1.25$ and 10 in dimensions 2, 3, 4, 5, 8, 9, 10, 12, and 20. Each run used a recorded deterministic seed. The complete results are reported in Table 1.

All 18 theoretical values lie within the corresponding two-sided normal 95% Monte Carlo intervals: the maximum absolute standardized discrepancy is 1.660, and the maximum absolute relative error is 2.113×10^{-4} . The largest absolute quadrature error estimate reported by the adaptive routine across these cases is 3.552×10^{-11} . These results provide numerical support for the theoretical predictions.

Table 1

Monte Carlo validation of the stable quadrature formula. Each row uses 10^8 samples. Here $\varepsilon_{\text{rel}} = (\hat{a}^{\text{MC}} - a^{\text{quad}})/a^{\text{quad}}$, and z is the discrepancy divided by its Monte Carlo standard error. The notation xek denotes $x \times 10^k$.

$\alpha = 1.25$						$\alpha = 10$					
d	a^{quad}	$\hat{a}^{\text{MC}}$	s.e.	$10^4 \varepsilon_{\text{rel}}$	z	d	a^{quad}	$\hat{a}^{\text{MC}}$	s.e.	$10^4 \varepsilon_{\text{rel}}$	z
2	0.41429435	0.41431538	2.501e-5	0.508	0.841	2	1.19666680	1.19677769	6.732e-5	0.910	1.617
3	0.15209303	0.15209032	1.226e-5	-0.178	-0.221	3	1.2306930	1.2306571	9.406e-5	-0.291	-0.381
4	0.04995978	0.04995401	4.721e-6	-1.156	-1.223	4	1.1257499	1.1255807	1.019e-4	-1.503	-1.660
5	0.01493763	0.01493994	1.572e-6	1.546	1.469	5	0.93604703	0.93620155	9.501e-5	1.651	1.626
8	2.567962e-4	2.567629e-4	3.299e-8	-1.297	-1.009	8	0.34611889	0.34607030	4.336e-5	-1.404	-1.120
9	5.884669e-5	5.885745e-5	7.912e-9	1.827	1.359	9	0.22068237	0.22068795	2.900e-5	0.253	0.192
10	1.283446e-5	1.283558e-5	1.795e-9	0.869	0.622	10	0.13394539	0.13397369	1.835e-5	2.113	1.542
12	5.352895e-7	5.352658e-7	8.002e-11	-0.443	-0.297	12	0.04328974	0.04329731	6.355e-6	1.749	1.192
20	4.188191e-13	4.187867e-13	7.450e-17	-0.772	-0.434	20	1.226485e-4	1.226420e-4	2.155e-8	-0.527	-0.300

7. Conclusion

We determined the volume of the stepping-stone diversion neighborhood in $\mathbb{R}^d$ for every $d \geq 2$. For two distinct sites p and q , the volume factorizes as

$$\lambda_d(S_{\text{SS},\alpha}(p, q)) = \|p - q\|^d a_{d,\text{SS}}(\alpha), \quad (93)$$

where the normalized constant depends only on d and α . Using an explicit boundary parametrization and orthogonal cross-sections, we reduced the computation of $a_{d,\text{SS}}(\alpha)$ to a one-dimensional integral valid for every $\alpha > 1$.

The formula recovers the degenerate and Gabriel region cases. At the other endpoint, we strengthen the previously recognized planar graph limit: in every dimension, the finite-parameter neighborhoods, after removal of the defining sites, increase from within to the open relative-neighborhood lune, and their volumes converge to Devroye's classical unit-lune constant. For each finite point set, the corresponding graph in fact stabilizes at the open-lune relative-neighborhood graph for a finite parameter. Strict region inclusion makes the graphs decrease under edge inclusion, places them between the relative-neighborhood and Gabriel graphs in the corresponding parameter ranges, and yields connectivity for every nonempty finite point set and every $\alpha \geq 1$. The parameter $\alpha = 2$ is also the sharp threshold for guaranteed geometric and abstract planarity in dimension two. For $\alpha \geq 2$, Gabriel or Delaunay edges provide candidate sets, but these inclusions do not by themselves give a linear-time planar algorithm or a subquadratic higher-dimensional algorithm.

Devroye's regular-set theorem identifies the limiting conditional expected degree in every dimension and yields a general asymptotic lower bound for the expected number of edges:

$$\lim_{n \rightarrow \infty} \mathbb{E}[N_n(X_1) \mid X_1 = x] = \frac{\kappa_d}{a_{d,\text{SS}}(\alpha)}. \quad (94)$$

For planar graphs with $\alpha \geq 2$, worst-case sparsity upgrades that bound to the exact limit $\pi/[2a_{2,\text{SS}}(\alpha)]$ under Devroye's density regularity condition. In Watanabe's planar homogeneous Poisson point process, it gives the corresponding graph-edge-length density

$$2\rho a_{2,\text{SS}}(\alpha) r e^{-\rho a_{2,\text{SS}}(\alpha) r^2}, \quad (95)$$

which specializes to Watanabe's Gabriel formula at $\alpha = 2$ and approaches his relative-neighborhood formula as $\alpha \rightarrow \infty$. This calculation and the planar volume formula are planar specializations. The Delaunay containment itself, the general Devroye degree limit and edge-count lower bound, and the remaining geometric, analytic, and graph-theoretic results hold for every $d \geq 2$.

A change of variables yields stable quadrature, independently validated by a rejection sampler applied to the original region inequality. Across 18 runs of 10^8 samples, every discrepancy was below two Monte Carlo standard errors. Continuity and strict monotonicity additionally make the normalized volume, and hence the limiting expected degree, an invertible calibration of the diversion parameter.

CRediT authorship contribution statement

Heriberto Espino-Montelongo: Conceptualization, Methodology, Formal analysis, Investigation, Validation, Visualization, Writing – original draft. **Héctor Maravillo:** Investigation, Supervision, Visualization, Writing – review and editing.

Code and data availability

No external datasets were used. The exact computational release is archived on Zenodo under DOI [10.5281/zenodo.21291269](https://doi.org/10.5281/zenodo.21291269) (Espino-Montelongo and Maravillo, 2026). It contains the quadrature, Monte Carlo, and figure-generation code, outputs, random seeds, and reproduction instructions. The public GitHub repository hosts the maintained version of these materials. To complement Figure 1, the same companion provides an interactive three-dimensional visualization of the normalized diversion neighborhood; the parameter α can be varied continuously, and the region can be rotated and magnified. Two supplementary CSV files underlying Table 1 add the hit counts, acceptance rates, sampling-cylinder volumes, quadrature error estimates, sample counts, and random seeds for each run.

Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation, the authors used ChatGPT by OpenAI for language, better wording, drafting, $\LaTeX$ table creation and the idea on Section 6. Authors reviewed and verified all resulting material and take full responsibility for the manuscript. All figures were generated deterministically from mathematical formulas and numerical data using author-verified code; no generative image model was used.

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